

# Aiden Aquino

703.743.4357 | [aquinoax@dukes.jmu.edu](mailto:aquinoax@dukes.jmu.edu) | [Github](#) | [LinkedIn](#)

## EDUCATION

### James Madison University

2023 - Current

*B.S. in Computer Science, Minor in Mathematics and Data Analytics*

- GPA: 3.851 | Presidents List (Fall 2023, Spring 2025, Spring 2026), Deans List (Spring 2024, Fall 2024)
- **Relevant Computer Science Coursework:** Algorithms & Data Structures, Software Engineering, Computer Systems I & II, Discrete Structures I & II, Programming Languages
- **Relevant Mathematics Coursework:** Probability & Statistics, Linear Algebra, Calculus II

## EXPERIENCE

### Undergraduate Researcher - James Madison University Mathematics and Statistics

Spring 2026 - Present

*AI-Powered Digital Twin for Water Distribution System Cybersecurity*

- Developed AI-powered digital twin for industrial control system cybersecurity using SWaT and BATADAL datasets.
- Designed and implemented anomaly detection models (LSTM autoencoders, Isolation Forest) for identifying attacks in OT environments.
- Built data pipelines and evaluation workflows for reproducible testing of ICS security models.
- Conducted system-level blast-radius analysis of cyberattack propagation and simulation-based resilience evaluation for cyberattack response.

### Teaching Assistant - James Madison University

*Lead Teaching Assistant*

Fall 2026 - Incoming

- Mentor and support teaching assistants in delivering course instruction and student assistance.
- Coordinate with instructors to improve course operations and student support workflows.
- Train and onboard new teaching assistants each semester.

*Teaching Assistant*

Fall 2025 - Present

- Conduct weekly lab hours to support 30+ students in programming assignments and debugging.
- Provide one-on-one and small-group instruction on course concepts.

## PROJECTS

### Box Programming Language (Interpreter & Compiler)

- Designed and implemented a custom programming language with lexer, parser, AST interpreter, and scoped runtime environment.
- Implemented recursive descent parsing with operator precedence and expression evaluation through AST traversal.
- Developed language runtime supporting variables, functions, control flow, and nested scope execution.
- Added structured error handling with source-position tracking for syntax and runtime diagnostics.

### Mini-ELF Binary Analysis Tool

- Developed a command-line tool to parse Mini-ELF/Y86 binary files, inspect memory structure, and disassemble code and data sections.
- Simulated instruction-level execution with debug tracing to analyze runtime behavior.

### Recipe Management System

- Developed desktop GUI application for managing recipes, meals, and inventory.
- Designed structured data model for recipe and inventory management with persistent storage.

## SKILLS

**Languages:** Java, Python, C#, Ruby, C, Rust, Haskell

**Tools:** Git, GitHub, Linux, Docker, SQL

**Libraries/Frameworks/Databases:** FastAPI, React, Pandas, NumPy, scikit-learn, PyTorch, TimescaleDB